

7 Troubleshooting/Service

7.1 Overview

Introduction

Instrument performance can be adversely affected by installation and application problems as well as by hardware problems. We recommend that you investigate the problems in the following order:

- installation related problems
- application related problems
- hardware and software related problems

and use the information presented in this section to solve them.

What's in this section?

The following topics are covered in this section.

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Installation related problems

Read the Installation section in this manual to make sure the UDC3200 has been properly installed. The installation section provides information on protection against electrical noise, connecting external equipment to the controller, and shielding and routing external wiring.

ATTENTION System noise induced into the controller will result in diagnostic error messages recurring. If the diagnostic error messages can be cleared, it indicates a “soft” failure and is probably noise related.

If system noise is suspected, completely isolate the controller from all field wiring. Use calibration sources to simulate PV and check all controller functions; i.e. Gain, Rate, Reset, Output, Alarms, etc.

Application related problems

Review the application of the controller; then, if necessary, direct your questions to the local sales office.

Hardware and software related problems

Use the troubleshooting error message prompts and controller failure symptoms to identify typical failures which may occur in the controller. Follow the troubleshooting procedures to correct them.

7.2 Troubleshooting Aids

Overall error messages

An error message can occur:

- At power-up. See Subsection 7.3.
- When the Status Tests are requested. See Subsection 7.4.
- During continuous background tests while in normal operation. See Subsection 7.5.

Controller failure symptoms

Other failures may occur that deal with the Power, Output, or Alarms. Refer to the controller failure symptom in Table 7-4 to determine what is wrong and the troubleshooting procedures to use to correct the problem.

Check installation

If a set of symptoms still persists, refer to *Section 2 - Installation* and ensure proper installation and proper use of the controller in the system.

Customer support

If you cannot solve the problem using the troubleshooting procedures listed in this section, you can get **technical assistance** by dialing 1-800-423-9883 USA and Canada.

An engineer will discuss your problem with you. **Please have your complete model number, serial number, and Software version available.** The model and serial numbers can be found on the chassis nameplate. The software version can be viewed under Setup Group "Status." See Table 7-1.

If it is determined that a hardware problem exists, a replacement controller or part will be shipped with instructions for returning the defective unit.

Do not return your controller without authorization from Honeywell's Technical Assistance Center or until the replacement has been received.

Check out Honeywell's **web site** at <http://www.honeywell.com/imc>.

Determining the software version

Table 7-1 lists the procedure for identifying the software version number.

Table 7-1 Procedure for Identifying the Software Version

Step	Operation	Press	Result
1	Select STATUS Set Up Group		<i>Upper Display = READ</i> <i>Lower Display = STATUS</i>
2	Read the software version		You will see: <i>Upper Display = Software version number</i> 32xx <i>Lower Display = VERSION</i> Please give this number to the Customer Support person. It will indicate which version of UDC3200 you have and help them determine a solution to your problem.

7.3 Power-up Tests

What happens at power-up

When power is applied, the controller will run three diagnostic tests. After these tests are completed, “TEST DONE” is displayed.

Failsafe Failures

If one or more of these tests fail, the controller will go to the Failsafe Manual Mode, and FAILSF will flash in the lower display and a message indicating which test failed will appear in the lower display. Then, “DONE” will appear in the lower display.

Three Position Step test failures

For controller configured for Three Position Step Control with motor position indication and Auto-cal has never been done, the prompt CAL MTR will appear to suggest that the controller be calibrated. *Refer to Subsection 6.4 - Three Position Step Output Calibration.*

7.4 Status Tests

Introduction

When required, the results of these tests can be checked to determine the reason the controller has gone to Failsafe.

How to check the status tests

The procedure in Table 7-2 tells you how to display the results of the status tests.

Table 7-2 Procedure for Displaying the Status Test Results

Step	Operation	Press	Result
1	Select STATUS Set Up Group		Upper Display = READ Lower Display = STATUS
2	Read the test results		You will see: Upper Display = NO or YES YES indicates a failure Lower Display = FAILSAFE
			Upper Display = PASS or FAIL Lower Display = TEST

7.5 Background Tests

Introduction

The UDC3200 performs ongoing background tests to verify data and memory integrity. If there is a malfunction, a diagnostic message will be displayed (blinking) in the lower display.

In the case of simultaneous malfunctions, the messages will appear in sequence in the lower display. Table 7-3 lists these background tests, the reason for their failure, and how to correct the problem.

Diagnostic messages may be suppressed (stop the blinking) by pressing the RUN/HOLD key. The messages will still be available for viewing by pressing the LOWER DISPLAY key.

Table 7-3 Background Tests

Lower Display	Reason for Failure	How to Correct the Problem
EE FAIL	Unable to write to non-volatile memory. Anytime you change a parameter and it is not accepted, you will see E FAIL.	1. Check the accuracy of the parameter and re-enter. 2. Try to change something in configuration. 3. Run through Read STATUS tests to re-write to EEPROM.
FAILSAFE	This error message shows whenever the controller goes into a failsafe mode of operation. This will happen if: • RAM test failed • Configuration test failed • Calibration test failed • Burnout configured for none and the input failed.	1. Run through STATUS check to determine the reason for the failure. 2. Press the SET UP key until STATUS appears in the lower display. 3. Press the FUNCTION key to see whether the tests pass or fail, then run through the STATUS codes a second time to see if the error cleared.
INP1 RNG	Input 1 out of range. The process input is outside the range limits.	1. Make sure the range and actuation are configured properly. 2. Check the input source. 3. Restore the factory calibration. (See <i>Subsection 5.6.</i>) 4. Field calibrate. See <i>Section 5 - Input Calibration.</i>
INP1FAIL	Two consecutive failures of input 1 integration; i.e., cannot make analog to digital conversion. This will happen if: • Upscale or Downscale burnout is selected and the input is open • Input not configured correctly for the sensor being used • Input source is grossly out of range.	1. Make sure the actuation is configured correctly. See <i>Section 3 - Configuration.</i> 2. Make sure the input is correct and that it has not burned-out (opened). 3. Check for gross over-ranging with a multimeter. 4. Restore factory calibration. See <i>Subsection 5.6.</i>
INP2 RNG	Input 2 out of range. The remote input is outside the range limits.	Same as INP1RNG above.
INP2FAIL	Two consecutive failures of input 2 integration. i.e., cannot make analog to digital conversion.	Same as INP1FAIL above.

Lower Display	Reason for Failure	How to Correct the Problem
CONF ERR	• PV low limit is > PV high limit • SP low limit is > SP high limit • Output low limit > Output high limit	1. Check the configuration for each item and reconfigure if necessary.
PV LIMIT	PV out of range. $PV = INP1 \times RATIO1 + INP1 \text{ BIAS}$	1. Make sure the input signal is correct. 2. Make sure the Ratio and Bias settings are correct. 3. Recheck the calibration. Use Bias of 0.0
RV LIMIT	The result of the formula shown below is beyond the range of the remote variable. $RV = INP2 \times RATIO + BIAS$	1. Make sure the input signal is correct. 2. Make sure the Ratio2 and Bias2 settings are correct. 3. Recheck the calibration. Use a Ratio2 of 1.0 and a Bias2 of 0.0.
SEGERR	Setpoint Program start segment number is less than ending segment number.	Check SP Program configuration, subsection 3.5 Set up Group SPPROG function prompts "STRSEG" and "ENDSEG".
CAL MTR	Slidewire calibration never performed.	Field Calibrate the slidewire. See Section 6.4.
SW FAIL	Position Proportional slidewire input failure.	See the Trouble Shooting Procedure in Section 7.7.
SOOTING	Percent Carbon falls outside sooting boundary	Check process for correct operation.
TCx WARN	Thermocouple on Input x (1 or 2) is starting to burnout.	This diagnostic message means that the controller has detected that the thermocouple is starting to burnout. This error message may also be created if the resistance of the wires used to connect the thermocouple to the instrument is above 100 ohms.
TCxFAIL	The Thermocouple on Input x (1 or 2) is in imminent danger of burning out.	This diagnostic message means that the controller has detected that the thermocouple will soon fail. User should consider replacing the thermocouple as soon as possible. This message will also be generated if the resistance of the wires used to connect the thermocouple to the instrument is above 180 ohms.
CURFAIL	Current Output is less than 3.5 mA.	The current output is open circuit. Check the field wiring. See Procedure #2.
AXFAIL	Auxiliary Output is less than 3.5 mA.	The auxiliary output is open circuit. Check the field wiring. See Procedure #9.

7.6 Controller Failure Symptoms

Introduction

In addition to the error message prompts, there are failure symptoms that can be identified by noting how the controller displays and indicators are reacting.

Symptoms

Compare your symptoms with those shown in Table 7-4.

Table 7-4 Controller Failure Symptoms

Upper Display	Lower Display	Indicators	Controller Output	Probable Cause	Trouble-shooting Procedure
Upper Display	Lower Display	Indicators	Controller Output	Probable Cause	Trouble-shooting Procedure
Blank	Blank	Off	None	Power Failure	1
OK	Displayed Output disagrees with Controller Output	OK	Controller Output disagrees with Displayed Output	Current Proportional Output	2
OK		OK		Position Proportional or TPSC Output	3
OK		OK		Time Proportional Output	4
OK		OK		Current/Time Proportional Output	5
OK	OK	OK	External Alarm function does not operate properly	Malfunction in alarm output	6
Display does not change when a key is pressed				Keyboard Malfunction	7
Controller fails to go into "Slave" operation during communications				Communications Failure	8
OK	Displayed Output disagrees with Auxiliary Output	OK	Controller Auxiliary Output disagrees with Displayed Auxiliary Output	Auxiliary Output	9

Other symptoms

If a set of symptoms or prompts other than the one you started with appears while troubleshooting, re-evaluate the symptoms. This may lead to a different troubleshooting procedure.

If the symptom still persists, refer to the installation section in this manual to ensure proper installation and proper use of the controller in your system.

7.7 Troubleshooting Procedures

Introduction

The troubleshooting procedures are listed in numerical order as they appear in Table 7-4. Each procedure lists what to do if you have that particular failure and how to do it or where to find the data needed to accomplish the task.



WARNING—SHOCK HAZARD

TROUBLESHOOTING MAY REQUIRE ACCESS TO HAZARDOUS LIVE CIRCUITS, AND SHOULD ONLY BE PERFORMED BY QUALIFIED SERVICE PERSONNEL. MORE THAN ONE SWITCH MAY BE REQUIRED TO DE-ENERGIZE UNIT BEFORE SERVICING.

Equipment needed

You will need the following equipment in order to troubleshoot the symptoms listed in the tables that follow:

- Multimeter – Capable of measuring millivolts, milliamps and resistance.
- Calibration sources – T/C, mV, Volt, etc.

Procedure #1

Table 7-5 explains how to troubleshoot power failure symptoms.

Table 7-5 Troubleshooting Power Failure Symptoms

Step	What to do	How to do it
1	Check the AC line voltage.	Use a voltmeter to measure the AC voltage across terminals L1 and L2 on the rear terminal panel of the controller. Check the earth ground connection.
2	Make sure the chassis plugs into the rear of the case properly.	Withdraw the chassis and visually inspect the controller board and the inside of the case.
3	Check the system for Brown-outs, heavy load switching, etc., and conformance to installation instructions.	Refer to <i>Section 2 - Installation</i> .
4	Change Power board.	Installation instructions supplied with new board.

Procedure #2

Table 7-6 explains how to troubleshoot Current Output failure symptoms.

Table 7-6 Troubleshooting Current Output Failure

Step	What to do	How to do it
1	Make sure the controller is configured for Current output and the proper range (4 to 20 or 0 to 20) is configured.	Make Output Set Up group function prompt OUT ALG = CUR. Make the Output Set UP group function prompt CRANGE = 4–20 or 0–20 per your application. Refer to <i>Section 3 - Configuration</i> .
2	Check the field wiring.	Output impedance must be less than or equal to 1000 ohms.
3	Check the output.	Put the controller into Manual mode and change the output from 0 % to 100 % (4-20 mA). Use a DC milliammeter at the rear terminals to verify the output.
4	Recalibrate the Current Proportional output.	Refer to <i>Section 6 - Output Calibration</i> for details.
5	Change Current Output board.	Installation instructions provided with new board.
6	Change Controller	

Procedure #3

Table 7-7 explains how to troubleshoot Position Proportional Output failure symptoms.

Table 7-7 Troubleshooting Position Proportional Output Failure

Step	What to do	How to do it
1	Make certain that the controller is configured for Position Proportional output.	Make Output Algorithm Set Up group function prompt OUT ALG = POSITON. Refer to <i>Section 3 - Configuration</i> .
2	Check the field wiring.	Refer to <i>Section 2 - Installation</i> for details.
3	Check the output.	Put the controller into Manual mode and change the output from 0 % to 100 %.
4	Check whether the motor drives in both directions. If it does go to Step 6.	See the Position Proportional calibration procedure in <i>Section 6 - Output Calibration</i> for motor slidewire calibration.
5	Check whether the motor drives in either direction. If the motor drives only in one direction, check the slidewire. If the motor does not drive in either direction, check the motor.	Refer to the motor instructions.
6	Check the output voltage to the slidewire (if used).	See the Input 2 wiring diagram for terminal designations in <i>Section 2 – Installation</i> . 1) Measure between the R and the – terminals (across the entire slidewire). This voltage should be between 0.024 volts (for a 100 ohm slidewire) and 0.200 volts (for a 1000 ohm slidewire). This voltage should remain steady as the motor moves. 2) Measure between the + and the – terminals (across just the wiper section). This voltage should be somewhere between 0.002 and 0.200 volts, depending upon the total slidewire resistance and the current position of the slidewire. This voltage should vary smoothly as the motor moves and the wiper travels across the slidewire. If the voltage jumps at a particular position while the motor is moving, this could indicate a “deadspot” and mean that the slidewire may be worn out and need replacing. Refer to the motor manufacturer’s instructions.

7	Make sure the output relays are actuating properly.	Put the controller into Manual mode. Vary the output above and below the present value. Observe "OUT" indicators and the output value ("OUT") on the lower display. When the "OUT 1" indicator is on, the output value should be increasing. When the "OUT 2" indicator is on, the output value should be decreasing. If these are not working properly, check the field wiring, then go to Step 5. If they are, go to Step 8.
8	Recalibrate the controller.	Refer to <i>Section 6 - Output Calibration</i> .
9	Change the two Output Relays or the Dual Relay Board (depending upon unit)	Installation instructions supplied with the new relays or board.

Procedure #4

Table 7-8 explains how to troubleshoot Time Proportional Output failure.

Table 7-8 Troubleshooting Time Proportional Output Failure

Step	What to do	How to do it
1	Make sure the controller is configured for Time Proportional output.	Make Output Algorithm Set Up group function prompt OUTALG = RLY or RLYD. Refer to <i>Section 3 - Configuration</i>.
2	Check the field wiring.	Make sure the NO or NC contact wiring is correct. Refer to <i>Section 2 - Installation</i> for details.
3	Check the output.	Put the controller into Manual mode. Vary the output above and below the present value. Observe OUT1 indicator on the operator interface. Contact should change state. 0 % open, 100 % closed. Listen for a click from the relay when the OUT1 indicator changes state.
4	Check relay.	Change relay.
5	Change MCU board.	Installation instructions supplied with the new board.

Procedure #5

Table 7-9 explains how to troubleshoot Current/Time or Time/Current Proportional Output failure.

Table 7-9 Troubleshooting Current/Time or Time/Current Proportional Output Failure

Step	What to do	How to do it
1	Make sure the controller is configured for Time/Current or Current/Time Proportional output.	Make Output Algorithm Set Up group function prompt OUT ALG = TCUR or CURT. Refer to <i>Section 3 – Configuration</i> .
2	Check the field wiring.	Make sure the NO or NC contact wiring selection is correct. Refer to <i>Section 2 - Installation</i> for details.
3	Check the relay output.	Put the controller into Manual mode. Vary the output above and below the present value. Observe OUT1 indicator on the operator interface. Listen for a click from the relay when the OUT1 indicator changes state.
4	Check the Current Proportional Output.	Put the controller into Manual mode and change the output from 0 % to 100 % (4-20 mA). Use a DC milliammeter at the rear terminals to verify the output.
5	Recalibrate the controller.	Refer to <i>Section 6 - Output Calibration</i> for details.
6	Change MCU and/or Current Output boards.	Installation instructions supplied with new board.

Procedure #6

Table 7-10 explains how to troubleshoot Alarm Relay Output failure.

Table 7-10 Troubleshooting Alarm Relay Output Failure

Step	What to do	How to do it
1	Check the alarm configuration data. If it is correct, check the field wiring.	Reconfigure if necessary. Refer to <i>Section 3 - Configuration</i> for details.
2	Check that the applicable alarm relay actuates properly depending on what you have set at prompt AxSxTYPE. If it does, check the field wiring.	If the alarm type is set for PV, place the controller in manual mode. Vary the input to raise and lower the PV around the setpoint. Listen for a click from the relay as the PV moves in either direction and note that the proper ALM1 or ALM2 is lit. EXAMPLE: If the alarm is set for MAN, put the controller into manual mode. The alarm light is ON. Put the controller into automatic mode and the alarm light is OFF.
3	Check the contacts.	Make sure the NO or NC contact wiring is correct. Refer to <i>Section 2 - Installation</i> for relay contact information.
4	Change the relay and/or the current output board.	Installation instructions supplied with the new relay or board.
5	Change MCU board.	Installation instructions supplied with the new board.

Procedure #7

Table 7-11 explains how to troubleshoot a Keyboard failure.

Table 7-11 Troubleshooting a Keyboard Failure

Step	What to do	How to do it
1	Make sure the keyboard is connected properly to the MCU/output and power/input boards.	Withdraw the chassis from the case and visually inspect the connection.
2	Controller Keyboard or specific keys may be LOCKED OUT via the security code.	Use your four-digit security code number to change the lockout level. Refer to <i>Section 3 – Configuration</i> .
3	Run the keyboard test.	Press the [SET UP] key and hold in, then press the [FUNCTION] key at the same time. The controller will run a display test. Then you will see: Upper Display KEYS Lower Display TRY ALL Press each key. If it works, the key name will appear in the lower display.
4	Replace the display/keyboard if any keys do not function.	Refer to <i>“Parts Replacement Procedures”</i> in this section.

Procedure #8

Table 7-11 explains how to troubleshoot a RS 485 Communications failure.

Table 7-12 Troubleshooting a RS-485 Communications Failure

Step	What to do	How to do it
1	Check the Address Number, ComState and Baud Rate settings.	See Section 3.13.
2	Check the field wiring and termination resistor.	Using an ohm meter, check the resistance across the communications rear terminals. See Section 2.7 for wiring diagrams.
3	Make sure the Communications Printed Wiring Board is installed properly in the controller.	Withdraw the chassis from the case and inspect the board. See the exploded view (Figure 8-1) for location of the board. Return the chassis to the case.
4	Determine if the Communications board is faulty by running a LOCAL LOOPBACK TEST. If the test fails, replace the board. If the test passes, the problem is most likely elsewhere in the communications network.	Disconnect the communications cable from the rear terminals. Run the Local Loopback Test. Press [SET UP] until you see: Upper Display SET UP Lower Display COM Press [FUNCTION] until you see: Upper Display DISABLE Lower Display LOOPBACK Press  or  you will see: Upper Display ENABLE Lower Display LOOPBACK The test will run until the operator disables it here.

Procedure #9

Table 7-13 explains how to troubleshoot an Ethernet Communications failure.

Table 7-13 Troubleshooting an Ethernet Communications Failure

Step	What to do	How to do it
1	Check the IP address, Subnet Mask address and Gateway address settings.	See the PIE Tool Manual.
2	Check if the Ethernet Connection is active.	Looking into the instrument, there should be steady green LED. If this is not present, then the instrument is not seeing a valid Ethernet connection. See Section 2.7 for wiring diagrams. A second green LED will blink during actual Ethernet transactions.
3	Change Ethernet Communications board.	Installation instructions provided with new board.
4	Change Controller	

Procedure #10

Table 7-14 explains how to troubleshoot Auxiliary Proportional Output failure symptoms.

Table 7-14 Troubleshooting Auxiliary Output Failure

Step	What to do	How to do it
1	Make sure the controller is configured for Auxiliary Output and the proper range (4 to 20 or 0 to 20) is configured.	Make Options Set Up group function prompt AUX OUT any selection other than NONE. If this prompt does not show up, check if DIG IN 2 is enabled. If so, then as Auxiliary Output and Digital Input 2 are mutually exclusive, you must choose which one of these features you wish to use. Make the Options Set UP group function prompt CRANGE = 4–20 or 0–20 per your application. Refer to <i>Section 3 - Configuration</i>.
2	Check the field wiring.	Output impedance must be less than or equal to 1000 ohms.
3	Check the output.	Change the AUX OUT selection to OUTPUT. Put the controller into Manual mode and change the output from 0 % to 100 % (4-20 mA). Use a DC milliammeter at the rear terminals to verify the output.
4	Recalibrate the Auxiliary output.	Refer to <i>Section 6 - Output Calibration</i> for details.
5	Change Auxiliary Output board.	Installation instructions provided with new board.
6	Change Controller	

7.8 Restoring Factory Configuration

Introduction

This procedure restores the configuration of the instrument back to the Factory Settings per Section 3.16.

ATTENTION: Restoring the factory configuration overwrites all user-entered configuration changes. This procedure cannot be undone, it is a one-way process.

Table 7-15 explains how to restore Factory Configuration.

Table 7-15 Restoring Factory Configuration

Step	What to do
1	Turn off the power to the instrument for at least five seconds.
2	Turn the power back on and simultaneously press the “FUNCTION” and  keys. This must be done while “TEST DONE” is being displayed.
3	If step 2 was performed correctly, the instrument will now display “UDC” [Upper] “UPDATE” [Lower].
4	Press the Function Key. The instrument will now display “DIS” [Upper] “RESTORE” [Lower].
5	Press the  key. The instrument will now display “CONFIG” [Upper] “RESTORE” [Lower].
6	Press the Function Key. The instrument will now display “DOIN” “RESTORE”
7	When the instrument finishes the restore operation, it automatically resets itself and restarts in the product mode. The instrument configuration will now be the same as it was when the instrument left the factory and all user-entered configurations since that time have been overwritten.

7.9 Software Upgrades

Introduction

This procedure enables software features that were not ordered from the factory. See Table 8-3 for a list of the available Software Upgrades.

ATTENTION:

This procedure cannot be undone, it is a one-way process.

Each instrument has a unique code number sequence, so the following procedure must be performed on each instrument to be upgraded.

Table 7-16 explains how to enable new software features.

Table 7-16 Software Upgrades

Step	What to do
1	Turn off the power to the instrument for at least five seconds.
2	Turn the power back on and simultaneously press the FUNCTION and  keys. This must be done while “TEST DONE” is being displayed.
3	If step 2 was performed correctly, the instrument will now display “UDC” [Upper] “UPDATE” [Lower].
4	Press the Function Key. The instrument will now display DISABLE [Upper] “RESTORE” [Lower].
5	Press the  key. The instrument will now display “CONFIG” [Upper] “RESTORE” [Lower].
6	Press the  Key. The instrument will now display OPTIONS [Upper] “RESTORE” [Lower].
7	Press the FUNCTION Key. The instrument will now display “XXXX” [Upper] “ENTER1” [Lower], where XXXX is a unique code number for this particular instrument. Write this number down.
8	Press the FUNCTION Key. The instrument will now display “XXXX” “ENTER2”. Write this number down.
9	Press the FUNCTION Key. The instrument will now display “XXXX” “ENTER3”. Write this number down.
10	Write down the Model and Serial Numbers of your instrument.

- 11 Contact your Honeywell Representative to place an order. Please have a company purchase order number available before you call. The order entry person will ask for the following information:
1. Software Upgrade Part Number you require:
Math Software Upgrade – 50004635-501, or
Set Point Programming Software Upgrade – 50004635-502
 2. Model Number of your instrument(s)
 3. Serial Number of your instrument(s)
 4. Code Numbers 1, 2 and 3 from your instrument(s)
 5. Purchase order number.

With this information, a new code number set will be generated for your instrument.

- 12 When you have the new code number set, repeat steps 1 to 6.
- 13 Press the **FUNCTION** Key. The instrument will now display “XXXX” “ENTER1”, where XXXX is a unique code number for this particular instrument. Using the ▲ and ▼ keys, enter the new Code 1 number.
- 14 Press the **FUNCTION** Key. The instrument will now display “XXXX” “ENTER2”. Using the ▲ and ▼ keys, enter the new Code 2 number.
- 15 Press the **FUNCTION** Key. The instrument will now display “XXXX” “ENTER3”. Using the ▲ and ▼ keys, enter the new Code 3 number.
- 16 Press the **FUNCTION** Key. The instrument will process the new code numbers and add the new software feature. If the code numbers were entered incorrectly or if the wrong code numbers for this particular instrument were entered, then the controller will go into Manual Mode and flash the message “FAILSAFE” on the lower display. Check the code numbers being entered and repeat steps 12 through 16.
- 17 When the instrument finishes the operation, it automatically resets itself and restarts in the product mode. The instrument configuration now includes the added software feature(s).
-